

JBM9204 INTRODUCTION TO BIOMETRIC SYSTEMS

-Open Elective

CLASS - III AI & ML

SEMESTER - V

REGULATION 2023



UNIT-1 BIOMETRIC SYSTEMS AND AUTHENTICATION TECHNOLOGIES

Unit-1 Biometric Systems And Authentication Technologies

COURSE OBJECTIVE: To familiarize the fundamental principles of different biometric systems.

SYLLABUS

Biometrics – Authentication vs. Verification, Biometric characteristics - Types of Biometrics – General Architecture of Biometric System and Functionalities – Biometric applications - Authentication technologies – Need for strong authentication – Protecting privacy and biometrics policy – Performance measures.

COURSE OUTCOME 1: Apply basic language processing techniques using regular expressions, automata, morphology, and n-gram models.

BIOMETRICS

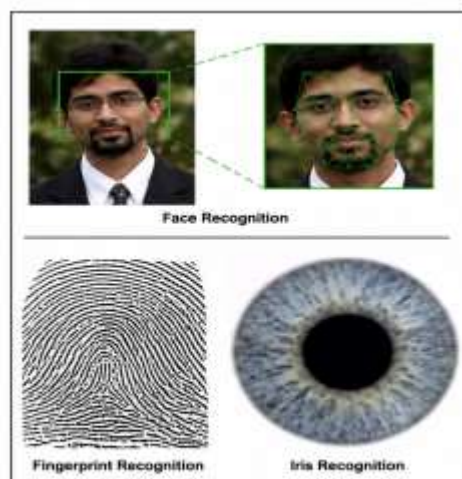
- **Definition:** The measurement and statistical analysis of a person's unique physical and behavioral characteristics - primarily used for identification, security, and access control, allowing our body (or actions) to act as an inherent password.
- It provides **high security** by using traits that are difficult to duplicate or steal.
- Traditional methods of establishing a person's identity → include knowledge-based (e.g., passwords) and token-based (e.g., ID cards) mechanisms → but these surrogate representations of the identity can easily be lost, shared, manipulated or stolen thereby undermining the intended security.
- Biometrics → offers a natural and reliable solution to certain aspects of identity management by utilizing fully automated or semi-automated schemes to recognize individuals based on their inherent physical and/or behavioral characteristics → possible to establish an identity based on "who you are" rather than by "what you possess" or "what you remember".

BIOMETRICS



(a)

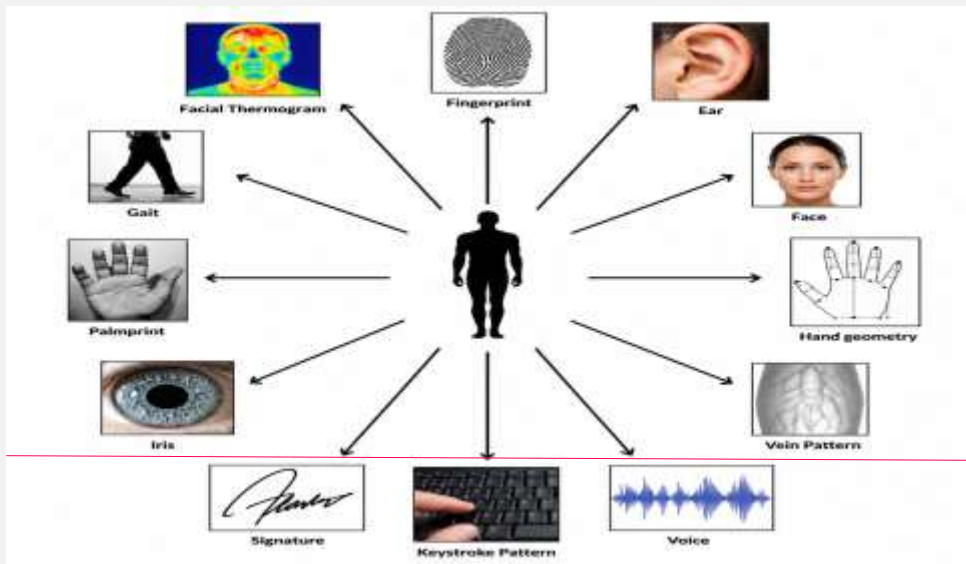
(a) Traditional Methods



(b)

(b) Biometrics

BIOMETRICS



Examples of Biometric Traits

BIOMETRICS

- Required in applications that render services to only legitimately enrolled users (like sharing networked computer resources, granting access to nuclear facilities, performing remote financial transactions or boarding a commercial flight.)
- Widely used in authentication (confirming a claimed identity) and identification (determining an unknown identity).
- Generally consist of data acquisition, feature extraction, template storage, matching, and decision-making modules.
- Commonly used in smartphones, banking, border security, healthcare, attendance systems, Aadhaar, e-passports, and access control.
- Challenges → privacy concerns, template security, spoofing attacks, environmental variations, and system cost.

Authentication vs. Verification

Aspect	Authentication	Verification
Definition	Validates that the returning user is indeed the same person trying to access a system (via a password, biometrics, or an OTP)	Confirms a person's identity and background against authoritative sources (like a government ID) to build trust
What it is	The process of proving your real-world identity to a system	The process of confirming that the person trying to log in is the exact same verified individual who registered the account
In short..	Proves you are still that same person	Proves who you are
Primary Goal	To confirm the legitimacy of a user's claimed identity.	To protect access by ensuring the user is authorized to enter a system.

Authentication vs. Verification

Aspect	Authentication	Verification
When It Happens	Usually a one-time process during initial onboarding or registration.	An ongoing process that occurs every time a user logs in or accesses a secure page, or performs a sensitive action
How it works	You submit official proof (e.g., uploading a passport, scanning a driver's license, or taking a live selfie to prove you are a real person)	You provide pre-arranged digital credentials, such as a password, biometric scan (like a fingerprint), or a One-Time Password (OTP).
Methods Used	Government-issued IDs (passports, driver's licenses), liveness checks, and address proofs.	Credentials like passwords, PINs, biometrics (FaceID), and one-time passcodes (OTP).
Data Evaluated	Cross-referenced against external/authoritative databases and document issuers.	Compared against internal credentials stored securely by the system.

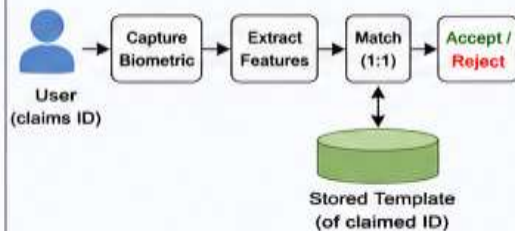
Authentication vs. Verification

1. AUTHENTICATION vs. VERIFICATION PROCESS

AUTHENTICATION

(1 : 1 Comparison)

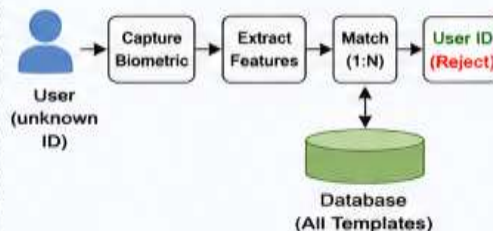
The system verifies a claimed identity by comparing the biometric of the user with the stored template of that claimed identity.



VERIFICATION (IDENTIFICATION)

(1 : N Comparison)

The system identifies the user by comparing the biometric with all templates in the database and returns the best match.

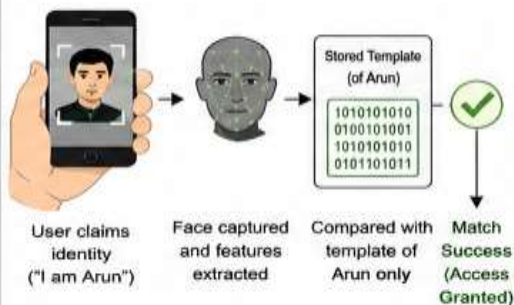


Authentication vs. Verification – An Example

AUTHENTICATION vs. VERIFICATION EXAMPLE

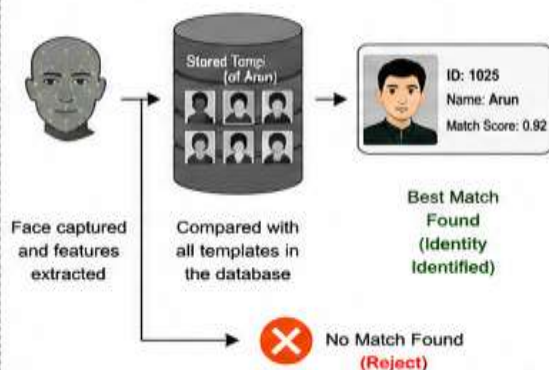
AUTHENTICATION (1:1)

Example: Logging into a smartphone



VERIFICATION / IDENTIFICATION (1:N)

Example: Finding a person in a database



Biometric Characteristics

Factors that determine the suitability of a physical or a behavioral trait to be used in a biometric application:

S. No.	Factor	Description
1	Universality	Every individual accessing the application should possess the trait.
2	Uniqueness	Sufficiently different trait across individuals comprising the population.
3	Permanence	Trait is sufficiently invariant over a period of time with respect to the matching algorithm. A trait that changes significantly over time is not a useful biometric.
4	Measurability	Should be possible to acquire and digitize the biometric trait using suitable devices that do not cause undue inconvenience to the individual → the acquired raw data should be open to processing in order to extract representative feature sets.

Biometric Characteristics

Factors that determine the suitability of a physical or a behavioral trait to be used in a biometric application:

S. No.	Factor	Description
5	Performance	The recognition accuracy and the resources required to achieve that accuracy should meet the constraints imposed by the application
6	Acceptability	Individuals in the target population that will utilize the application should be willing to present their biometric trait to the system
7	Circumvention (by-passing)	The ease with which the trait of an individual can be imitated using artifacts (e.g., fake fingers), in the case of physical traits, and mimicry, in the case of behavioral traits.



Biometric Characteristics

- Biometric characteristics are the unique, measurable biological and behavioral traits used to identify or authenticate an individual. Because they are tied directly to a person, they offer a highly secure, convenient alternative to traditional passwords or ID cards.
- A **single biometric** cannot effectively meet all the requirements (e.g., accuracy, practicality, cost) imposed by all applications → hence a number of them are admissible.
- The **relevance** of a specific biometric to an application is established depending upon the **nature and requirements of the application**, and the properties of the biometric characteristic.
- Commonly used biometric characteristics:

1. Face	4. Palmprint	7. Signature
2. Fingerprint	5. Iris	8. Voice
3. Hand geometry	6. Keystroke	9. Gait



Biometric Characteristics

I. Face

Face recognition identifies a person using facial features such as the eyes, nose, lips, chin, and their spatial relationships, or by analyzing the overall facial appearance.

Features

- Non-intrusive and natural method.
- Widely used in surveillance, smartphones, and access control.
- Can work without physical contact.

Advantages

- Easy to acquire images.
- Suitable for remote identification.
- Convenient for users.

Limitations

- Performance affected by:
 - ✓ Lighting conditions
 - ✓ Facial expressions
 - ✓ Pose variations
 - ✓ Aging
 - ✓ Background clutter
- Requires accurate face detection, localization, and recognition.



Biometric Characteristics

2. Fingerprint

Fingerprint recognition uses the unique ridge and valley patterns present on fingertips.

Features

- One of the oldest and most reliable biometric techniques.
- Even identical twins have different fingerprints.
- Uses minutiae features such as ridge endings and bifurcations.

Advantages

- Very high recognition accuracy.

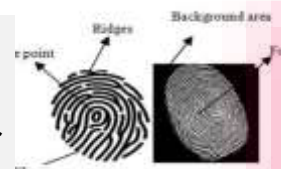
- Inexpensive fingerprint scanners.

- Widely used in:

- Aadhaar card
- Mobile phones
- Forensic applications
- Attendance systems

Limitations

- Damaged or worn fingerprints reduce accuracy.
- Manual labor may affect fingerprint quality.
- Large databases require computational resources.



Biometric Characteristics

3. Hand geometry

Hand geometry measures the physical dimensions of the hand, including:

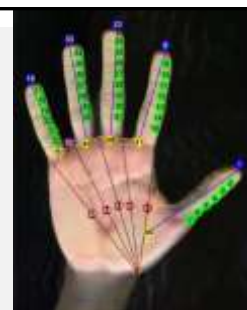
- ✓ Palm size
- ✓ Finger length
- ✓ Finger width
- ✓ Hand shape

Advantages

- Simple and easy to use.
- Low-cost implementation.
- Less affected by skin conditions.

Limitations

- Not highly distinctive.
- Unsuitable for identifying millions of users.
- Hand size changes in children.
- Jewelry or arthritis may affect measurements.
- Devices are relatively bulky



Biometric Characteristics

4. Palmprint

Palmprint recognition analyzes the ridge patterns, principal lines, wrinkles, and texture of the human palm.

Features

- Larger surface area than fingerprints.
- Can combine multiple palm features for improved accuracy

Advantages

- High distinctiveness.
- Better accuracy than fingerprint alone.
- Lower-resolution images can still capture principal lines

Limitations

- Sensors are larger and more expensive.
- Requires a larger image acquisition area.



Biometric Characteristics

5. Iris

Iris → the colored ring surrounding the pupil. Its complex texture is unique for every individual.

Features

- Highly distinctive.
- Stable throughout life.
- Even identical twins have different iris patterns.

Advantages

- Extremely high accuracy.
- Very low False Accept Rate (FAR).
- Suitable for large-scale identification.
- Can detect fake iris contact lenses

Limitations

- Higher False Reject Rate (FRR).
- Earlier systems were expensive.
- Requires user cooperation and good image quality



Biometric Characteristics

6. Keystroke

Keystroke dynamics identifies users based on their typing rhythm and speed.

Features

- Behavioral biometric.
- Measures timing between key presses and releases.

Advantages

- No additional hardware required.

- Can continuously verify users after login.
- Useful for computer security.

Limitations

- Typing style changes due to:
 - Stress
 - Fatigue
 - Keyboard type
 - Emotional state
- Not sufficiently unique for large-scale identification.



Biometric Characteristics

7. Signature

Signature recognition verifies a person's identity based on the way they sign their name.

Features

- Behavioral biometric.
- Online systems capture:
 - Speed
 - Pressure
 - Stroke order
 - Pen movement

Advantages

- Socially accepted for legal and banking transactions.
- Suitable for tablets and digital devices.

Limitations

- Signatures vary over time.
- Emotional and physical conditions affect signatures.
- Skilled forgeries may deceive the system.



Biometric Characteristics

8. Voice

Voice recognition identifies individuals based on vocal characteristics.

Features

- Combination of physiological and behavioral traits.
- Can be:
 - Text-dependent
 - Text-independent

Advantages

- Convenient and hands-free.
- Suitable for telephone and remote authentication.
- No special hardware required

Limitations

- Affected by:
 - Background noise
 - Illness
 - Emotional condition
 - Communication channel quality
- Less suitable for large-scale identification.



Biometric Characteristics

9. Gait

Gait recognition identifies individuals based on their walking style.

Features

- Uses body movement and walking patterns.
- Suitable for long-distance surveillance.

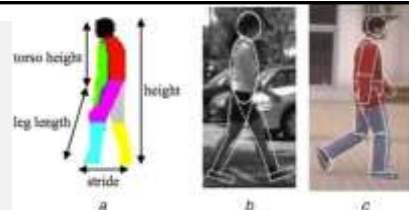
Advantages

- Non-contact biometric.

- Can identify people from a distance.
- Useful for security and surveillance

Limitations

- Influenced by:
 - Footwear
 - Clothing
 - Injuries
 - Walking surface
- Lower recognition accuracy than fingerprints or iris



Biometric Trait	Type	Major Advantage	Major Limitation
Face	Physiological	Non-contact and user-friendly	Sensitive to lighting and pose
Fingerprint	Physiological	High accuracy and low cost	Damaged fingerprints reduce accuracy
Hand Geometry	Physiological	Simple and inexpensive	Low uniqueness
Palmprint	Physiological	Rich feature information	Large sensors required
Iris	Physiological	Very high accuracy	Expensive and requires cooperation
Keystroke	Behavioral	No extra hardware	Typing pattern varies
Signature	Behavioral	Widely accepted legally	Easy to forge and varies over time
Voice	Behavioral + Physiological	Convenient for remote access	Affected by noise and illness
Gait	Behavioral	Distance recognition	Influenced by clothing and walking style

Types of Biometrics

1. Physiological biometrics

Identify a person based on physical characteristics that are generally permanent and unique → these characteristics remain relatively stable throughout a person's lifetime.

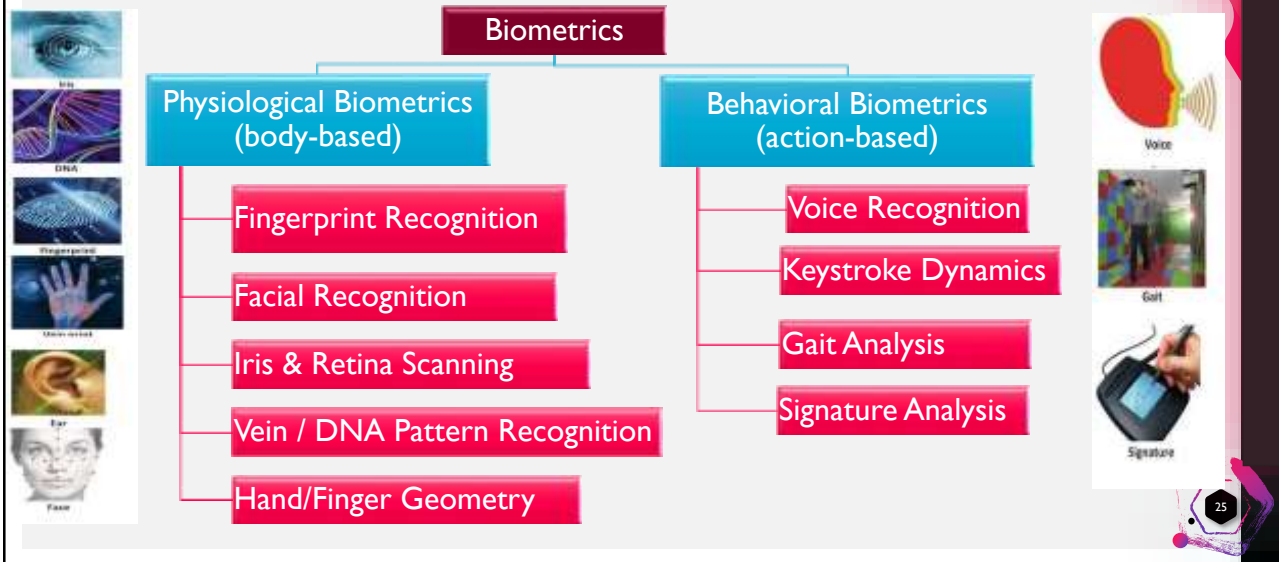
- Based on Physical characteristics
- Highly stable
- High accuracy
- User cooperation is usually required
- Eg. Fingerprint, Face, Iris, Palmprint

2. Behavioral biometrics

Recognize individuals based on behavioral patterns or habits developed over time → these traits may change due to age, health, or emotional state.

- Based on human behavior
- Stability may change over time
- Moderate accuracy
- User interruption or conscious is often less
- Eg. Signature, Voice, Gait, Keystroke

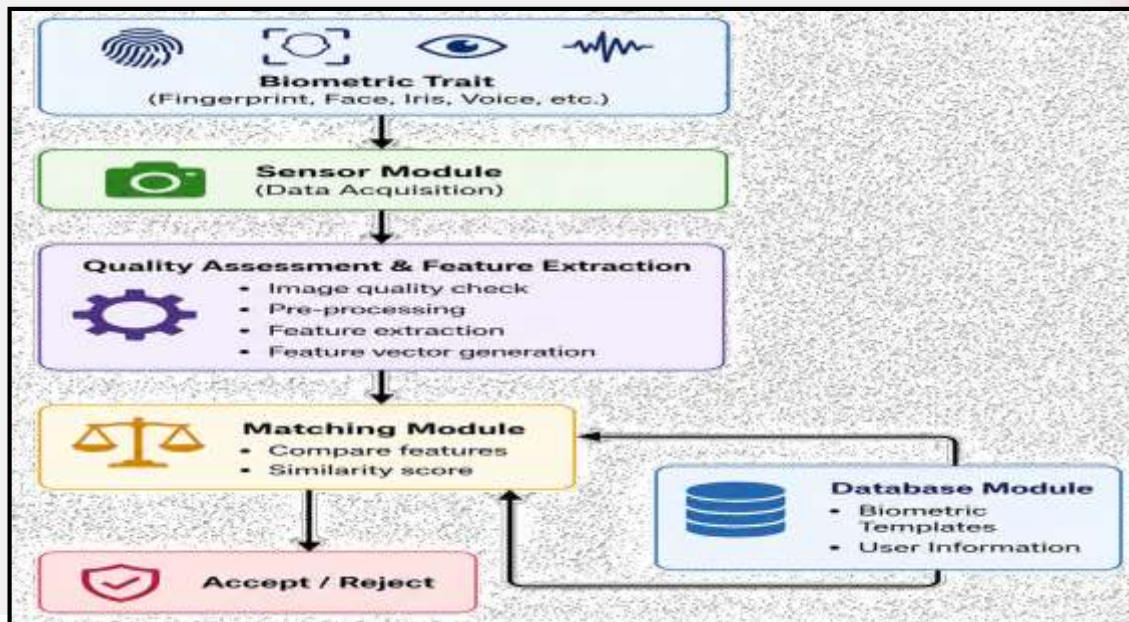
Types of Biometrics



General Architecture of Biometric System and Functionalities

- A biometric system → a pattern recognition system that
 - **Acquires** biometric data from an individual,
 - **Extracts** a salient feature set from the data,
 - **Compares** this feature set against the feature set(s) stored in the database,
 - **Executes** an action based on the result of the comparison.
- A generic biometric system has four main modules:
 1. A sensor module
 2. A quality assessment and feature extraction module
 3. A matching module and
 4. A database module.

General Architecture of Biometric System



General Architecture of Biometric System and Functionalities

Sensor Module

- Responsible for capturing the **raw biometric data** of a user using a suitable biometric sensor or scanner (e.g., fingerprint scanner, iris scanner, face camera, microphone).
- Acts as the **human-machine interface**, making it one of the most important components of a biometric system.
- The **quality** of the captured data depends on the sensor technology used.
- A poorly designed sensor may result in a **high Failure-to-Acquire (FTA)** rate, reducing system accuracy and user acceptance.

General Architecture of Biometric System and Functionalities

Quality Assessment and Feature Extraction Module

- After acquisition, the biometric sample is **evaluated** for quality to ensure it is suitable for processing.
- If the quality is **poor**, the user may be asked to provide the biometric sample again (**re-acquisition**).
- The data is then enhanced and processed to extract distinctive / **unique features** (e.g., fingerprint minutiae, iris patterns, facial landmarks).
- During enrollment, these extracted features are stored as **a biometric template** in the database for future authentication.
- A biometric template → a secure mathematical representation of unique physical or behavioral traits



General Architecture of Biometric System and Functionalities

Matching Module

- The module **compares** the extracted biometric features with the stored templates in the database and computes a matching score.
- The score indicates the **similarity** between the input sample and the stored template.
- Based on this score, the Decision-Making Module either **verifies** a claimed identity (1:1) or **identifies** the user (1:N) by selecting the best matching template.
- The quality of the biometric sample may also influence the matching score.



General Architecture of Biometric System and Functionalities

Database Module

- The module serves as the **repository** of biometric templates and user information.
- **Templates** are created during the enrollment process.
- Stored templates (multiple) provided to the matching module.
- During enrollment, the extracted biometric features (template) are stored along with user details such as name, PIN, or address.
- Depending on the application, enrollment may be
 - Supervised (e.g., bank ATM registration) or
 - Unsupervised (e.g., creating a biometric login on a personal computer).
- Raw biometric images stored in the database → gallery images
- Images acquired as input during authentication → probe images.



General Architecture of Biometric System and Functionalities

Module	Function
Sensor Module	Captures the biometric sample (fingerprint, face, iris, voice, etc.).
Quality Assessment & Feature Extraction Module	Checks sample quality, preprocesses it, and extracts unique biometric features to form a template.
Matching Module	Compares the extracted features with stored templates and computes a matching score.
Database Module	Stores enrolled biometric templates and provides them to the matching module during authentication or identification.

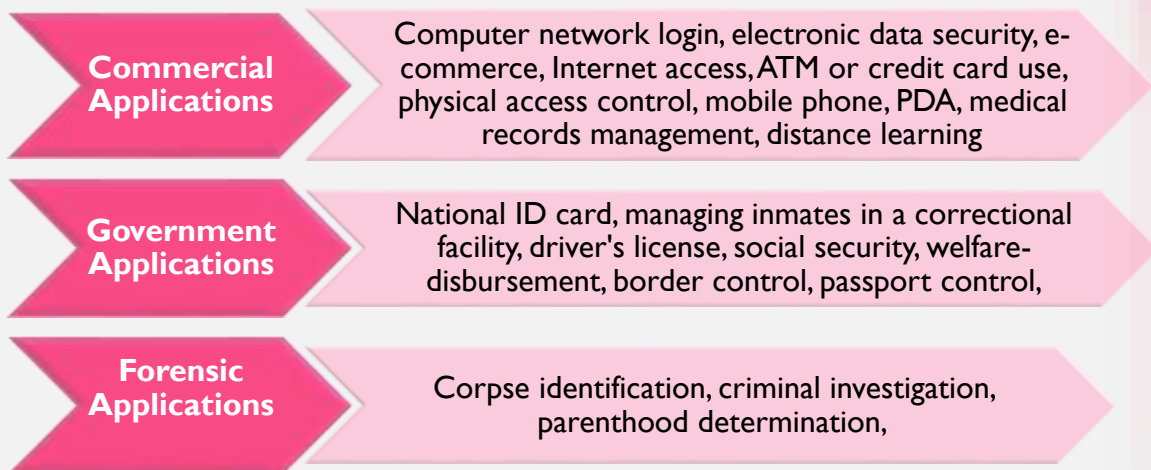


Biometric Applications

- Variety of scenarios ranging from issuing a driver's license to gaining entry into a country pose few questions on identity of a person, like
 - "Is he / she really who he / she claims to be?"
 - "Is this person authorized to use / access this facility?"
 - "Is he in the watch-list posted by the government?"
- This implies the increase in need for reliable user authentication techniques in the wake of higher concerns about security, and rapid advancements in networking, communication and mobility.
- Hence biometrics is being increasingly incorporated in several different applications.
- Applications categorized into 3 main groups. → Commercial applications, Government applications, Forensic applications



Biometric Applications - Categories



Biometric Applications - Examples

1. Access Control

- Secures entry to buildings, offices, laboratories, and data centers.
- Uses fingerprint, face, iris, or palm vein recognition instead of keys or access cards.
- **Example:** Fingerprint-based access to research laboratories.

2. Mobile Device Authentication

- Unlocks smartphones, tablets, and laptops.
- Replaces PINs and passwords.
- **Example:** Face ID, Fingerprint unlock, Windows Hello



Biometric Applications - Examples

3. Banking and Financial Services

- Secure login to online banking.
- ATM authentication.
- Mobile payment authorization.
- Fraud prevention.
- **Example:** Fingerprint authentication for UPI and mobile banking apps.

4. National Identity Systems

- Unique identification of citizens.
- Elimination of duplicate identities.
- Efficient delivery of government services.
- **Example:** **Aadhaar** in India uses fingerprints, iris scans, and facial images.



Biometric Applications - Examples

5. Border Control and Immigration

- Automated passport verification.
- E-passports.
- Immigration clearance.
- **Example:** Face and fingerprint verification at international airports.

6. Attendance Management

- Employee attendance.
- Student attendance.
- Prevents proxy attendance.
- **Example:** Fingerprint attendance system in colleges.



Biometric Applications - Examples

7. Law Enforcement and Criminal Investigation

- Criminal identification.
- Suspect verification.
- Missing person identification.
- **Example:** Automated Fingerprint Identification System (AFIS).

8. Healthcare

- Patient identification.
- Prevents medical identity fraud.
- Secure access to electronic health records.
- **Example:** Iris authentication for accessing patient records.



Biometric Applications - Examples

9. E-Commerce and Digital Payments
10. Forensic Science
11. Smart Homes and IoT
12. Military and Defense
13. Vehicle Security
14. Workplace Security
15. Surveillance and Public Safety
16. Voting Systems
17. Travel and Hospitality



Biometric Applications – Real World Examples

- The Schiphol Privium scheme at Amsterdam's Schipol airport employs **iris scan smart cards** to speed up the immigration procedure.(tap your smart card to a biometric reader, then look into the camera scanner to verify your identity)
- The Ben Gurion International Airport at Tel Aviv employs **automated hand geometry-based identification kiosks** to enable Israeli citizens and frequent international travelers to rapidly go through the passport inspection process.
- Some financial institutions in Japan have installed **palm-vein authentication systems in their ATMs** to help validate the identity of a customer intending to conduct a transaction - A contactless sensor used.
- Kroger, a US supermarket chain, has deployed **fingerprint scanners** in some of its stores in order to help customers cash payroll checks or render payment after a purchase.
- Season pass holders accessing theme park facilities at Disney World, Orlando, have their **finger geometry information** stored in a central repository.
- The United States Visitor and Immigration Status Indicator Technology (USVISIT) is a **border security system** that has been deployed at 115 airports. Biometric entry and exit procedures applied.



Biometric Applications – Real World Examples



Biometric Applications – Real World Examples



Authentication Technologies

- **Authentication Technologies** → Security tools that **verify the identity** of a user, device, or application to grant access to digital systems.
- Modern methods secure environments by using a **combination of factors** to protect data:
 - knowledge (passwords)
 - possession (hardware tokens), and
 - inherence (biometrics)
- Few technologies
 1. Multi-Factor Authentication (MFA)
 2. Passwordless Authentication
 3. Biometric Authentication
 4. Adaptive & Risk-Based Authentication
 5. Single Sign-On (SSO)



Authentication Technologies

1. Multi-Factor Authentication (MFA)

- MFA requires users to provide two or more verification methods from different categories, drastically reducing unauthorized access.
- Very effective countermeasure to one of the main routes of compromise in use by threat actors.
- One-Time Passwords (OTP): Short, time-limited codes sent via SMS or generated by authenticator apps.
- Push Notifications: The system sends an approval prompt to a trusted mobile device (like a smartphone).

2. Passwordless Authentication

- This approach eliminates the need to remember traditional passwords.
- Instead, it uses cryptographic protocols (such as FIDO2 (Fast IDentity Online 2) security keys) or magic links sent to an authenticated email.



Authentication Technologies

3. Biometric Authentication

- Biometrics confirm identities based on unique biological or behavioral traits.
- Because of their accuracy, they are commonly used in mobile banking and secure facility access:
- Physical: Fingerprint scans, facial recognition, and iris or retina scans.
- Behavioral: Keystroke dynamics and voiceprint matching.

4. Adaptive & Risk-Based Authentication

- This technology evaluates contextual data (e.g., your IP address, device type, geographic location, and behavioral patterns) to determine the risk of a specific login attempt in real-time.



Authentication Technologies

5. Single Sign-On (SSO)

- SSO allows users to authenticate once using a primary set of credentials and gain access to multiple applications without having to log in separately for each one.
- Common framework protocols include Security Assertion Markup Language (SAML).



Need for Strong Authentication

- Authentication is verifying the identity of a user or device before granting access to a system or network. It is typically achieved by requiring users to provide some form of identification, such as a username and password.
- **Strong authentication** → a multi-layered security process requiring users to verify their identity using at least two independent factors (e.g., a password, a physical device, or biometrics) rather than relying on a single password.
- It prevents unauthorized access, protects sensitive data, and ensures compliance with industry regulations.



Need for Strong Authentication

The Role of Risk

- During a login request, the system assesses:
 - Locations. Where is the request coming from?
 - Timestamps. When is the user requesting a login?
 - Frequency. How often has the user tried to log in previously?
- Eg. Strong authentication process using an SMS One-time Passcode (OTP):

Step 1: Password

The person creates and memorizes a unique set of numbers and letters used to access the system.



Step 2: Possession

After typing in the correct password, a secondary string of letters and numbers is sent to the user's registered smartphone.



Step 3: Access

After tapping in the second set of details, the user can get into the system.



Need for Strong Authentication

Why Strong Authentication is Essential?

- Strong authentication is crucial for protecting against cyber attacks, particularly those that rely on stolen credentials.
- Cybercriminals are adept at devising new ways to steal login credentials, whether through phishing emails, social engineering tactics, or brute-force attacks.
- Once they have obtained valid credentials, they can gain unrestricted access to a system or network, putting sensitive data and resources at risk.
- Multi-factor authentication (MFA), can significantly reduce the risk of unauthorized access.



Need for Strong Authentication

Why Strong Authentication is Essential?

- **Protection Against Breaches:** Passwords alone can be easily phished, guessed, or stolen in data breaches. By requiring "something you have" (such as a hardware key) or "something you are" (like a fingerprint), attackers cannot access systems even if they steal your password.
- **Compliance Requirements:** Regulatory frameworks like the Payment Services Directive 2 (PSD2) for payments, GDPR, and HIPAA explicitly mandate multi-factor controls to ensure user privacy and data security.
- **Mitigating Human Error:** People often reuse weak passwords across multiple sites. Implementing Strong Authentication layers automatically reduces the risk of accidental exposure without relying on users to create complex credentials.



Need for Strong Authentication

Types of Strong Authentication

- Security questions.
- One-time passwords (OTPs)
- App-generated codes
- Specialized authentication apps
- Physical authentication keys
- Biometrics
- Cryptographic challenge response protocol



Need for Strong Authentication

Best Practices for Strong Authentication

- Use Multi-Factor Authentication
 - Effective way - can greatly reduce the risk of stolen credentials.
- Enforce Strong Password Policies
 - Forces users to use complex passwords and encourage them to use password managers to store their passwords securely
- Implement Biometric Authentication
 - Much harder to replicate than passwords and can provide additional protection
- Monitor and Analyze Authentication Logs
 - Analyzing authentication logs can also provide valuable insights into potential security vulnerabilities.



Protecting Privacy And Biometrics Policy

- Biometric privacy policies → protect users by mandating informed consent, data minimization, and strict encryption.
- Because compromised physical traits (fingerprints, faces) cannot be reset, regulations like the Indian Digital Personal Data Protection (DPDP) Act classify biometrics as highly sensitive, requiring secure local storage and limited retention.
- **Key practices with Biometric security and privacy policies :**
 - ✓ Explicit Consent & Transparency: Organizations must obtain written authorization and provide clear notices before capturing biometric data.
 - ✓ Data Minimization & Anonymity: Systems should store mathematical templates (irreversible numerical representations) rather than raw images, and only collect what is strictly necessary.
 - ✓ Strict Retention Limits: Data must be deleted immediately after its intended purpose is fulfilled (e.g., non-matches discarded immediately).
 - ✓ Robust Encryption: Protocols like end-to-end encryption prevent unauthorized access during data transfer and storage



Protecting Privacy And Biometrics Policy

Protecting Privacy in Biometrics

- Biometric data such as fingerprints, facial images, iris patterns, and voiceprints is unique and permanent. Since it cannot be changed like a password, protecting biometric data is essential to prevent identity theft and misuse.
- Few Privacy Protection Measures
 - Data Encryption
 - Template Storage
 - Data Minimization
 - Access Control
 - Anonymization and Tokenization



Protecting Privacy And Biometrics Policy

Protecting Privacy in Biometrics

1. Data Encryption

- Converts readable biometric data into ciphertext using encryption algorithms (e.g., AES, RSA).
- Only authorized users with the correct key can decrypt and access the data.
- Ensures confidentiality during storage and transmission.

2. Template Storage

- Stores a biometric template (mathematical representation) instead of the raw biometric image.
- Templates are used for authentication while reducing privacy risks.
- Whenever possible, templates are stored locally on the user's device (on-device processing).



Protecting Privacy And Biometrics Policy

Protecting Privacy in Biometrics

3. Data Minimization

- Collect only the biometric data required for a specific purpose.
- Store only essential biometric features.
- Retain data only for the required period.
- Use data only for its intended purpose.
- Prefer on-device processing to reduce exposure.

4. Access Control

- Restricts access to biometric data to authorized users and systems only.
- Prevents unauthorized viewing, modification, or misuse of sensitive biometric information.



Protecting Privacy And Biometrics Policy

Protecting Privacy in Biometrics

5. Anonymization and Tokenization

Anonymization: Removes personal identifiers so individuals cannot be identified from stored biometric data.

Tokenization: Replaces sensitive biometric information with a random token, protecting the user's identity while allowing secure data management.



Protecting Privacy And Biometrics Policy

Biometrics Policy

- A Biometrics Policy defines the rules for the collection, storage, processing, and use of biometric data to ensure security, privacy, legal compliance, and user trust.

Core Elements of a Biometrics Policy

- **Purpose of Collection** – Clearly specify why biometric data is collected.
- **User Consent** – Obtain informed consent before collecting biometric information.
- **Data Retention** – Define how long biometric data will be stored and when it will be securely deleted.
- **Legal Compliance** – Follow applicable data protection laws such as **GDPR, HIPAA, and India's DPDP Act**.
- **User Rights** – Allow users to access, update, or request deletion of their biometric data.
- **Breach Response Plan** – Establish procedures to respond quickly if biometric data is compromised.



Performance Measures

- **Password-based systems** - a **perfect match** between two alphanumeric strings is necessary in order to validate a user's identity.
- **A biometric system** - **rarely** encounters two samples of a user's biometric trait that result in exactly the **same feature set**.
- Reason → due to imperfect
 - **Sensing conditions** (e.g., noisy fingerprint due to sensor malfunction),
 - **Alterations** in the user's biometric characteristic (e.g., respiratory ailments impacting speaker recognition)
 - Changes in **ambient conditions** (e.g., inconsistent illumination levels in face recognition) and
 - Variations in the **user's interaction** with the sensor (e.g., occluded iris or partial fingerprints).
 - Two feature sets originating from the same biometric trait of a user look exactly the same very rarely → a perfect match between two might indicate the possibility of a **replay attack**.



Performance Measures

- **Intra-class variation** → The variability observed in the biometric feature set of an individual is → must be small
- **Inter-class variation** → the variability between feature sets originating from two different individuals → must be large.
- Small intra-class variation and large inter-class variation → useful feature set.
- The degree of similarity between two biometric feature sets is indicated by a similarity score
 - **Genuine or authentic score** → if the similarity score is a result of matching two samples of the same biometric trait of a user.
 - **Impostor score** → if the similarity score involves comparing two biometric samples originating from different users.



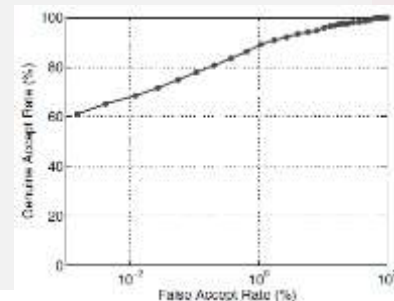
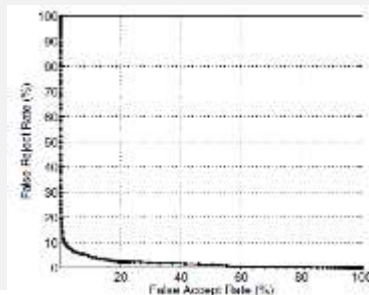
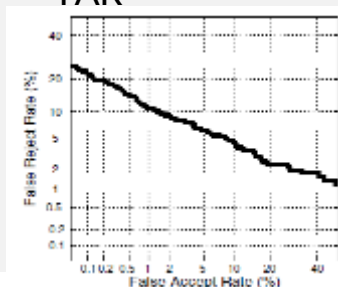
Performance Measures

- **False accept** $\rightarrow$ or false match $\rightarrow$ An impostor score that exceeds the threshold η .
- **False reject** $\rightarrow$ or false non-match $\rightarrow$ A genuine score that falls below the threshold η .
- The **False Accept Rate (FAR)** (or, the False Match Rate (FMR)) of a biometric system $\rightarrow$ defined as the fraction of impostor scores exceeding the threshold. η
- The **False Reject Rate (FRR)** (or, the False Non-match Rate (FNMR)) of a system $\rightarrow$ defined as the fraction of genuine scores falling below the threshold η .
- The **Genuine Accept Rate (GAR)** $\rightarrow$ the fraction of genuine scores exceeding the threshold η .
- **GAR = 1 - FRR**



Performance Measures

- **Detection Error Tradeoff (DET) curve** $\rightarrow$ plots the FRR against the FAR at various thresholds on a normal deviate scale and includes between these points.
- **Receiver Operating Characteristic (ROC) curve** $\rightarrow$ plots the FRR against the FAR at various thresholds on a linear, logarithmic or semi-logarithmic scale
- Generally, ROC curve plots the GAR (rather than the FRR) against the FAR



Performance Measures

- Doddington categorized biometric users into four categories of based on the inherent differences.
- It helps in understanding user behavior and improving system accuracy.

Category	Biometric Feature Sets	Intra-class / Inter-class Variations	FAR	FRR
Sheep	Very distinctive and safe users	Low intra-class High inter-class	Low	Low
Goats	Frequently rejected genuine users	Large intra-class	Low	High
Lambs	Easily impersonated users.	Low inter-class variations	High	Low
Wolves	Users who successfully impersonate others	Can imitate or manipulate biometric traits	High	Low

Performance Measures

- Examples:



Performance Measures

- **Equal Error Rate (EER)** $\rightarrow$ FAR = FRR $\rightarrow$ lower EER denotes better performance
- **Failure to Acquire (FTA) rate** (or, Failure to Capture (FTC) rate) $\rightarrow$ denotes the proportion of **times the biometric device fails** to capture a sample when the biometric characteristic is presented to it $\rightarrow$ the device is not able to locate a biometric signal of sufficiently good quality
 - Eg. an extremely faint fingerprint, a blocked face image).
 - Also impacted by sensor wear and tear.
 - Periodic sensor maintenance is helpful for the efficient functioning of a biometric system.
- **The Failure to Enroll (FTE) rate** $\rightarrow$ denotes the proportion of users that **cannot be successfully enrolled** in a biometric system.
 - Occurs when the system rejects poor quality inputs during enrollment
 - User training may be necessary in order to facilitate using the biometric system and the acquisition of good quality biometric data.



Performance Measures

Different errors

- False Accept Rate (FAR)
- False Reject Rate (FRR)
- Genuine Accept Rate (GAR)
- Equal Error Rate (EER) $\rightarrow$ FAR = FRR
- Failure to Acquire (FTA) rate
- Failure to Enroll (FTE) rate



Limitations of Biometric Systems

- Noise in sensed data
- Intra-class variations
- Inter-class similarities
- Non-universality
- Interoperability issues
- Spoof attacks



Thank
You